

SONGPAN, China

WMO: 561820

Lat: 32.6709N

Lon: 103.6030E

Elev: 2883

StdP: 71.16

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-11.2	-22.8	0.7	-1.0	-20.8	0.8	-3.3	6.6	3.5	5.6	1.6	1.5	0	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	25.1	13.8	23.7	13.3	22.2	12.8	15.3	22.1	14.7	20.9	14.1	19.8	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.3	13.7	16.8	12.7	13.1	16.5	12.1	12.6	16.4	54.5	22.2	52.3	21.0	50.3	19.8	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.1	4.3	-16.3	28.1	1.5	1.3	-17.4	29.0	-18.3	29.8	-19.2	30.5	-20.3	31.5		
(5)			-17.3	16.8	1.4	1.1	-18.3	17.6	-19.1	18.2	-19.9	18.9	-20.9	19.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-2.0	0.7	4.2	7.9	11.1	13.8	16.0	15.8	13.0	8.2	3.1	-1.2	(6)
(7)		DBStd	6.76	2.61	3.07	2.74	2.61	2.43	2.44	2.09	2.22	2.67	2.66	2.67	2.33	(7)
(8)		HDD10.0	1529	373	260	180	72	17	2	0	0	4	68	208	346	(8)
(9)		HDD18.3	3931	631	493	438	314	225	136	75	82	161	314	458	604	(9)
(10)		CDD10.0	651	0	0	1	8	50	117	187	181	95	12	0	0	(10)
(11)		CDD18.3	8	0	0	0	0	0	1	3	3	1	0	0	0	(11)
(12)		CDH23.3	156	0	0	0	1	2	14	63	63	12	0	0	0	(12)
(13)		CDH26.7	7	0	0	0	0	0	1	4	3	0	0	0	0	(13)
(14)	Wind	WSAvg	1.5	1.5	1.6	1.8	1.7	1.8	1.4	1.5	1.4	1.3	1.3	1.3	(14)	
(15)	Precipitation	PrecAvg	718	6	10	32	59	105	114	107	88	112	69	13	4	(15)
(16)		PrecMax	941	15	26	62	102	162	185	210	159	168	142	30	21	(16)
(17)		PrecMin	526	0	0	4	31	46	51	40	42	44	28	1	0	(17)
(18)		PrecStd	82	4	6	14	17	33	34	35	26	36	24	7	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	15.9	18.3	21.1	23.2	25.2	27.0	26.9	25.0	20.2	15.5	11.5	(19)
(20)			MCWB	1.1	3.1	5.0	7.7	10.7	13.4	14.3	14.8	13.9	10.6	4.6	1.4	(20)
(21)		2%	DB	9.3	12.6	15.3	18.7	20.8	22.7	25.0	25.1	22.6	17.4	13.4	9.1	(21)
(22)			MCWB	-0.1	1.9	4.2	7.1	10.1	12.7	14.1	13.8	12.8	8.6	3.8	0.2	(22)
(23)		5%	DB	7.1	10.4	13.2	16.6	18.9	21.0	23.5	23.6	20.5	15.4	11.5	7.4	(23)
(24)			MCWB	-1.1	1.1	3.5	6.5	9.4	12.2	13.7	13.5	11.9	8.0	3.1	-0.6	(24)
(25)		10%	DB	4.8	8.0	11.0	14.6	17.1	19.3	21.7	21.8	18.5	13.5	9.6	5.5	(25)
(26)			MCWB	-2.1	0.2	2.7	5.8	8.9	11.8	13.5	13.0	11.4	7.3	2.5	-1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	3.7	6.2	9.8	13.1	15.6	16.0	15.1	11.7	6.0	2.2	(27)	
(28)			MCDWB	10.5	13.2	15.9	17.8	20.2	22.1	23.4	24.4	22.6	18.5	13.4	10.2	(28)
(29)		2%	WB	0.1	2.5	5.1	8.3	11.5	14.4	15.3	15.1	13.9	10.2	4.6	0.7	(29)
(30)			MCDWB	8.3	11.4	13.0	15.6	17.7	20.0	22.0	22.2	20.2	15.5	11.3	7.9	(30)
(31)		5%	WB	-0.8	1.5	4.1	7.5	10.5	13.5	14.7	14.5	13.0	9.1	3.8	-0.3	(31)
(32)			MCDWB	6.3	9.4	11.4	14.5	16.4	18.7	20.8	20.9	18.5	13.7	10.0	6.5	(32)
(33)		10%	WB	-1.8	0.4	3.2	6.6	9.6	12.7	14.2	13.8	12.2	7.9	2.9	-1.2	(33)
(34)			MCDWB	4.4	7.3	9.7	13.0	15.1	17.4	19.6	19.7	17.1	12.3	8.4	5.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.5	15.0	13.3	12.6	11.6	10.5	11.4	11.7	10.6	10.5	13.4	15.0	(35)
(36)			MCDBR	19.3	18.9	17.5	16.9	15.3	13.9	15.0	15.2	14.3	13.6	16.6	17.9	(36)
(37)			MCWBR	11.1	9.8	8.0	7.1	6.1	5.2	5.5	5.5	5.7	6.3	8.6	10.1	(37)
(38)		5% WB	MCDBR	17.7	17.2	15.0	13.8	12.4	10.7	11.2	12.2	11.8	11.0	14.0	16.2	(38)
(39)			MCWBR	10.4	9.2	7.1	6.3	5.5	4.3	4.3	4.6	4.8	5.3	7.4	9.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.348	0.425	0.443	0.432	0.415	0.430	0.444	0.401	0.346	0.303	0.278	(40)	
(41)		taud	2.289	2.153	1.934	1.948	2.070	2.197	2.182	2.109	2.222	2.334	2.378	2.413	(41)	
(42)		Ebn at Noon	908	896	846	849	860	870	855	838	862	890	907	927	(42)	
(43)		Edh at Noon	105	134	180	186	166	146	148	156	134	110	95	88	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.44	3.99	4.68	5.15	5.07	4.55	4.94	4.66	3.81	3.41	3.38	3.23	(44)	
(45)		RadStd	0.23	0.34	0.27	0.36	0.32	0.49	0.58	0.55	0.34	0.34	0.24	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	-1.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.30	+0.29	-0.71	N/A	N/A	N/A	-34	-90	+75	+18

Nomenclature: See separate page